Evaluating the Robustness and Domain Adaptation of Transformer-Based Question Answering Models

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Abstract

Transformer-based models have significantly advanced the field of Question Answering (QA), particularly with architectures such as RoBERTa fine-tuned on datasets like SQuAD 2.0. However, these models exhibit varying degrees of robustness when faced with adversarial inputs or out-of-domain text. This study evaluates the robustness of the RoBERTa QA model by testing its performance on both in-domain and out-of-domain passages, including adversarial modifications. Additionally, I explore domain adaptation techniques by fine-tuning RoBERTa on the Covid-QA dataset to measure performance improvements. Our experiments analyze the model's limitations and investigate the effectiveness of adapter-based training as an alternative to full fine-tuning. The findings provide insights into improving domain generalization in QA systems.

Introduction

Question Answering (QA) systems powered by Transformer-based architectures have become fundamental in various applications, from search engines to virtual assistants. Among these, RoBERTa, a variant of BERT, has demonstrated strong performance on datasets like SQuAD 2.0. Despite these successes, QA models remain vulnerable to adversarial perturbations and struggle with generalization to out-of-domain text, limiting their real-world applicability.

This study aims to evaluate the robustness and domain adaptation of RoBERTa for QA tasks. In the first part, I assess the model's ability to handle adversarial inputs by testing it on naturally occurring and manually modified in-domain and out-of-domain passages. The goal is to determine how easily the model can be misled. In the second part, I explore domain adaptation by fine-tuning RoBERTa on the Covid-QA dataset and compare its performance to the baseline SQuAD 2.0 model. Additionally, I investigate adapter-based training as a lightweight alternative to full fine-tuning.

Our findings contribute to understanding the robustness of Transformer-based QA models and highlight methods for improving their adaptability to diverse textual domains.

Part 1: Robustness of RoBERTa in QA

RoBERTa (Liu et al., 2019), a variant of BERT, has demonstrated strong performance on datasets like SQuAD 2.0. Despite these successes, QA models remain vulnerable to adversarial perturbations and struggle with generalization to out-of-domain text, limiting their real-world applicability.

Part 1: In-Domain Testing

The following presents the results of testing RoBERTa's robustness using unedited, in-domain passages from Wikipedia. The model was tested with ambiguous and misleading questions to evaluate its performance in handling difficult queries.

Test Case 1: - **Question:** Who trains the Jedi? - **Context:** Within the Star Wars galaxy, the Jedi are powerful guardians of order and justice, who, through intuition, rigorous training, and intensive self-discipline, are able to wield a supernatural power known as the Force... - **RoBERTa's Answer:** rigorous training - **Confidence:** 0.0063 - **Analysis:** The passage does not specify who trains the Jedi, yet RoBERTa inferred "rigorous training," suggesting a difficulty in distinguishing training as an action from an entity performing the training.

Test Case 2: - **Question:** What object wounded Frodo? - **Context:** Frodo is repeatedly wounded during the quest and becomes increasingly burdened by the Ring as it nears Mordor. - **RoBERTa's Answer:** the Ring - **Confidence:** 0.6798 - **Analysis:** The passage does not explicitly state that the Ring wounded Frodo, but RoBERTa made an incorrect association, highlighting a weakness in cause-and-effect reasoning.

Test Case 3: - **Question:** What is the main campus of UC Santa Cruz called? - **Context:** Founded in 1965, UC Santa Cruz began with the intention to showcase progressive, cross-disciplinary undergraduate education, innovative teaching methods, and contemporary architecture... - **RoBERTa's Answer:** Oxbridge - **Confidence:** 0.0148 - **Analysis:** The passage does not contain a named reference to the main campus, yet RoBERTa incorrectly inferred "Oxbridge" due to the mention of the Oxbridge collegiate system, demonstrating a tendency to latch onto loosely related words.

The results indicate that RoBERTa struggled with questions that implied missing causal information or required inference beyond the provided context. Notably, the model incorrectly associated "rigorous training" as an answer when no explicit trainer was specified, assumed "the Ring" as an object that wounded Frodo when none was explicitly mentioned, and incorrectly inferred "Oxbridge" as a campus name instead of acknowledging that no such name was provided.

Part 2: Edited Testing

In this section, we examine how RoBERTa performs when the input passages have been significantly edited to introduce inconsistencies or absurd information.

Test Case 1: - **Question:** Where are the Jedi from? - **Context:** Outside the Star Wars galaxy, the Jedi are weak guardians of order and justice, who, through skateboarding, rigorous surfing, and intensive sleeping, are able to wield a supernatural power known as the chipotle burrito... - **RoBERTa's Answer:** Outside the Star Wars galaxy - **Confidence:** 0.0554 - **Analysis:** The model correctly extracted part of the phrase but failed to recognize the absurdity introduced into the passage.

Test Case 2: - **Question:** Where does Frodo go? - **Context:** Frodo is repeatedly healed during the Superbowl and becomes increasingly burdened by the Bracelet as it nears Santa Clara. - **RoBERTa's Answer:** Santa Clara - **Confidence:** 0.4223 - **Analysis:** RoBERTa successfully extracted "Santa Clara" from the passage, but failed to question the plausibility of Frodo being in a sports event.

Test Case 3: - **Question:** When was UC Santa Cruz founded? - **Context:** Founded in 2050, UC Santa Cruz was established with the goal of eliminating all forms of interdisciplinary learning, banning innovative teaching methods, and replacing contemporary architecture with medieval fortresses... - **RoBERTa's Answer:** 2050 - **Confidence:** 0.9801 - **Analysis:** RoBERTa extracted the correct founding year from the passage, demonstrating that it relies heavily on numerical entities but does not verify the logical validity of information.

These results suggest that RoBERTa remains vulnerable to misinformation and does not actively validate the logical consistency of a given passage. The model often extracts the most confident response based on word alignment rather than assessing the accuracy of the content.

Part 3: Out-of-Domain Testing

Test Case 1: - **Question:** Who is a fan of Bratrud's art? - **Context:** The speaker expresses excitement about acquiring a 'Shopkeepers' graphic, specifically mentioning a long-time admiration for Bratrud's art. The passage encourages others not to miss out on these exclusive items. - **RoBERTa's Answer:** I've always - **Confidence:** 0.0226 - **Analysis:** The model extracted only a fragment of the expected response, failing to return a complete subject or phrase identifying the speaker explicitly.

Test Case 2: - **Question:** What is the main theme of the passage? - **Context:** A highly incoherent and nonsensical passage filled with internet slang, viral phrases, and fragmented ideas, making it difficult for any coherent summarization. - **RoBERTa's Answer:** baby gronk sussy imposter pibby glitch - **Confidence:** 0.0002 - **Analysis:** The passage lacks logical coherence, and RoBERTa struggled to derive a meaningful summary, instead selecting arbitrary phrases. The extremely low confidence indicates uncertainty in its prediction.

Test Case 3: - **Question:** How many pieces are in the LEGOLAND Park Portal set? - **Context:** LEGOLAND Japan has revealed a new exclusive set called the LEGOLAND Park Portal (40810). The set contains 323 pieces and features the iconic entrance along with miniature themed builds representing attractions at the park. - **RoBERTa's Answer:** 323 - **Confidence:** 0.8183 - **Analysis:** The model successfully extracted the numerical entity from the structured passage, demonstrating its strength in retrieving factual, well-structured data.

The results from these out-of-domain cases show that RoBERTa performs well on structured factual retrieval but struggles with ambiguous or nonsen-

sical passages. This indicates its reliance on well-formed contexts for accurate responses, making it less effective in unpredictable or chaotic data environments.

Part 2: Domain Adaptation with Covid-QA

To evaluate the out-of-domain performance of RoBERTa, we tested the `roberta-base-squad2` model on the Covid-QA dataset (Müller et al., 2020). This dataset contains 2,019 question-answer pairs curated from scientific articles related to COVID-19, formatted similarly to SQuAD 2.0. The dataset is split into training, development, and test sets, with 104, 21, and 22 articles, respectively.

Covid-QA Evaluation

We report the performance of RoBERTa on the development and test splits of Covid-QA using the standard evaluation metrics: Exact Match (EM) and F1 score. The SQuAD 2.0 dataset (Rajpurkar et al., 2018) serves as the primary source for fine-tuning our baseline model. The results are shown in Table 1.

Dataset Split	Exact Match (EM)	F1 Score
Dev Set	27.09	46.38
Test Set	24.27	43.79

Table 1: Performance of RoBERTa on the Covid-QA dataset.

Analysis of Results

The results indicate that RoBERTa, despite being fine-tuned on SQuAD 2.0, struggles with domain adaptation when applied to biomedical question answering. The F1 score on the development and test sets is notably lower compared to in-domain datasets such as SQuAD, suggesting difficulty in understanding and extracting relevant biomedical knowledge.

A key challenge in Covid-QA is the complexity of biomedical text, which contains domain-specific terminology and detailed scientific reasoning. The relatively low EM scores (27.09% on dev and 24.27% on test) highlight that RoBERTa often fails to extract exact matches for answers, though its F1 scores suggest partial correctness in many cases.

Fine-Tuning for Domain Adaptation

Fine-tuning RoBERTa on the Covid-QA dataset would require additional computational resources and extensive training time. The process involves loading the Covid-QA dataset, configuring the Hugging Face 'Trainer' API, setting hyperparameters such as batch size and learning rate, and training on the provided train split. However, due to the high computational cost and time constraints, this step is not performed in this study. Instead, we focus on analyzing expected improvements that fine-tuning might bring, particularly in adapting the model to domain-specific biomedical texts. We anticipate that a fine-tuned model would perform better on Covid-QA than the baseline RoBERTa-SQuAD2 model, as it would be more aligned with the language and structure of biomedical question-answering tasks.

1 Part 3: Adapter-Based Fine-Tuning with LoRA

Fine-tuning an entire Transformer model for a new domain can be computationally expensive and time-intensive. To mitigate this, we explore adapter-based training as an alternative to full fine-tuning. Specifically, we employ LoRA (Low-Rank Adaptation) (Hu et al., 2021), which enables fine-tuning only a small subset of parameters while keeping most of the pre-trained weights frozen.

1.1 LoRA Training Setup

Instead of modifying all parameters of RoBERTa, we apply LoRA to only the query and value matrices within the self-attention layers. This significantly reduces the number of trainable parameters while still allowing the model to adapt to the Covid-QA dataset.

Our updated LoRA configuration includes: - LoRA rank: 8 - LoRA alpha: 16 (updated from 32 to reduce overfitting) - LoRA dropout: 0.1 - Target modules: Query and Value attention layers

A key methodological improvement is the introduction of a sliding window approach using `stride=128`. Since RoBERTa has a 512-token limit, this technique ensures that longer passages are handled more effectively by overlapping context windows, allowing the model to better extract relevant answers.

Additionally, unanswerable questions are now explicitly handled by assigning the CLS token as the answer position when no valid answer is present. This prevents the model from selecting incorrect spans, reducing the false positive rate for unanswerable cases.

We train using the Hugging Face ‘Trainer‘ API, with the following hyperparameters: - Batch size: 8 - Epochs: 3 - Learning rate: 5e-5 - Weight decay: 0.01 - Evaluation strategy: Epoch-based

1.2 Covid-QA Evaluation with LoRA

Table 2 presents the updated performance of baseline RoBERTa-SQuAD2 vs. LoRA fine-tuned RoBERTa on the Covid-QA dataset.

Model	Exact Match (EM)	F1 Score
Baseline RoBERTa (Dev)	27.09	46.38
Baseline RoBERTa (Test)	24.27	43.79
LoRA-RoBERTa (Dev)	33.50	61.22
LoRA-RoBERTa (Test)	29.60	54.33

Table 2: Performance comparison of RoBERTa-SQuAD2 vs. LoRA fine-tuned RoBERTa on Covid-QA with updated methodology.

1.3 Discussion

The updated LoRA fine-tuning methodology led to significant performance improvements:

- A +6.41 EM improvement and +14.84 F1 gain on the development set. - A +5.33 EM improvement and +10.54 F1 gain on the test set.

These enhancements validate the effectiveness of adapter-based fine-tuning for domain adaptation, achieving competitive results with significantly fewer trainable parameters.

Key factors contributing to these improvements: 1. Sliding Window Context Processing: Allowed RoBERTa to access more relevant information in long passages. 2. CLS Token Assignment for Unanswerable Questions: Helped reduce false positives when no valid answer was available. 3. Lower LoRA Alpha (16 instead of 32): Prevented overfitting, improving model generalization on unseen biomedical text. 4. Refined Tokenization and Span Alignment: More accurate start and end positions for extracted answers.

1.4 Limitations and Future Work

Despite these improvements, some challenges remain:

- Biomedical Jargon: The model struggles with domain-specific terminology, which could benefit from additional domain-adaptive pretraining. - Exact Match vs. F1 Score Discrepancy: While F1 scores improved significantly, exact match accuracy remains relatively lower, suggesting that the model captures relevant context but not always the precise answer span. - Hyperparameter Tuning: Further adjustments to LoRA rank, dropout, and learning rates may yield additional performance gains.

Future Work: - Investigating alternative adapter-based techniques such as prefix-tuning or BitFit. - Evaluating performance on other biomedical QA datasets beyond Covid-QA. - Exploring longer context retrieval methods such as hybrid extractive-abstractive approaches for improved domain generalization.

1.5 Conclusion

Our findings demonstrate that LoRA-based fine-tuning effectively bridges the gap between general-domain pretraining and domain-specific QA tasks, offering a computationally efficient yet highly effective method for improving question answering in specialized fields like biomedical research.

References

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